

4(a)	1 $q = \sqrt{0.11}$ or $q = 0.33166$ and $p = 1 - 0.33166$ or $p = 0.66834$; 2 $2pq = 2 \times 0.66834 \times 0.33166$ or $2pq = 0.44332$ and $\times 100$; 3 44 ; <i>max 1 if no working is shown but have correct answer</i>	3
4(b)	<i>apply ora throughout</i> 1 (only) continuous, is polygenetic / involves more than two genes / involves many genes ; 2 in continuous different alleles have a small(er) effect (on the trait) ; 3 (only) continuous has, genes / alleles, showing an additive effect ;	3
4(c)	any three from: 1 founder effect / genetic drift ; A description 2 geographical, isolation / separation / barrier (between populations) ; 3 no gene flow / no interbreeding / reproductive isolation (between populations) ; 4 different mutations occurred ; 5 different, selection pressures / environments ; 6 different changes in allele frequencies ;	3